

- 1 (a) gravitational force provides/is the centripetal force B1
 $GMm/r^2 = mv^2/r$ M1
 $v = \sqrt{GM/r}$ A0 [2]

allow gravitational field strength provides/is the centripetal acceleration (B1)
 $GM/r^2 = v^2/r$ (M1)

- (b) (i) kinetic energy increase/change = loss/change in (gravitational) potential energy B1
 $\frac{1}{2}mV_0^2 = GMm/x$ C1
 $V_0^2 = 2GM/x$
 $V_0 = \sqrt{2GM/x}$ A1 [3]

(max. 2 for use of r not x)

- (ii) V_0 is (always) greater than v (for $x = r$) M1
 so stone could not enter into orbit A1 [2]

(expressions in (a) and (b)(i) must be dimensionally correct)